Code No: 123CT **R15**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech II Year I Semester Examinations, September/October - 2023
DIGITAL LOGIC DESIGN
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART - A

(25 Marks)

1.a)	Realize 2-input AND gate using only 2-input NOR gates.	[2]
b)	Differentiate standard and canonical forms.	[3]
c)	What are the limitations of K-Map?	[2]
d)	Simplify the following using K-map. $f(A,B,C) = \prod(0,1,2,7)$	[3]
e)	Draw the half subtractor logic diagram.	[2]
f)	Differentiate combinational and sequential circuits.	[3]
g)	Write the characteristic equation for D flip-flop.	[2]
h)	Differentiate synchronous and asynchronous sequential circuits.	[3]
i)	What is meant by programmable logic array?	[2]
j)	Explain memory decoding.	[3]

PART - B

(50 Marks)

2. Discuss the theorems and postulates of Boolean algebra. [10]

OR

3. Express the following Boolean function in sum of minterms form.
$$f(A,B,C,D) = A + BC + D$$
Also, realize this using NAND gates. [10]

4. Simplify the following Boolean function and implement using only OR-AND logic.
$f(W,X,Y,Z) = \prod M(0,2,3,5,6,10,12,14)$ [10]

OR

5. Simplify the following Boolean function and implement using only AND-OR logic
$f(A,B,C,D) = \prod M(1,2,3,7,10,13,14)$ [10]

6. Implement the following Boolean function using 8 to 1 multiplexer.
$f(D,C,B,A) = \prod M(0,1,3,5,6,8,10,12,14,15)$ [10]

OR

7. Design a 3-bit binary to excess-3 code converter. [10]

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8. What is a shift register? Discuss different types of shift registers. [10]

OR

9. Design a 3-bit binary ripple counter. [10]

10. Compare all memories in detail. [10]

OR

11. Explain sequential memory in detail. [10]

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